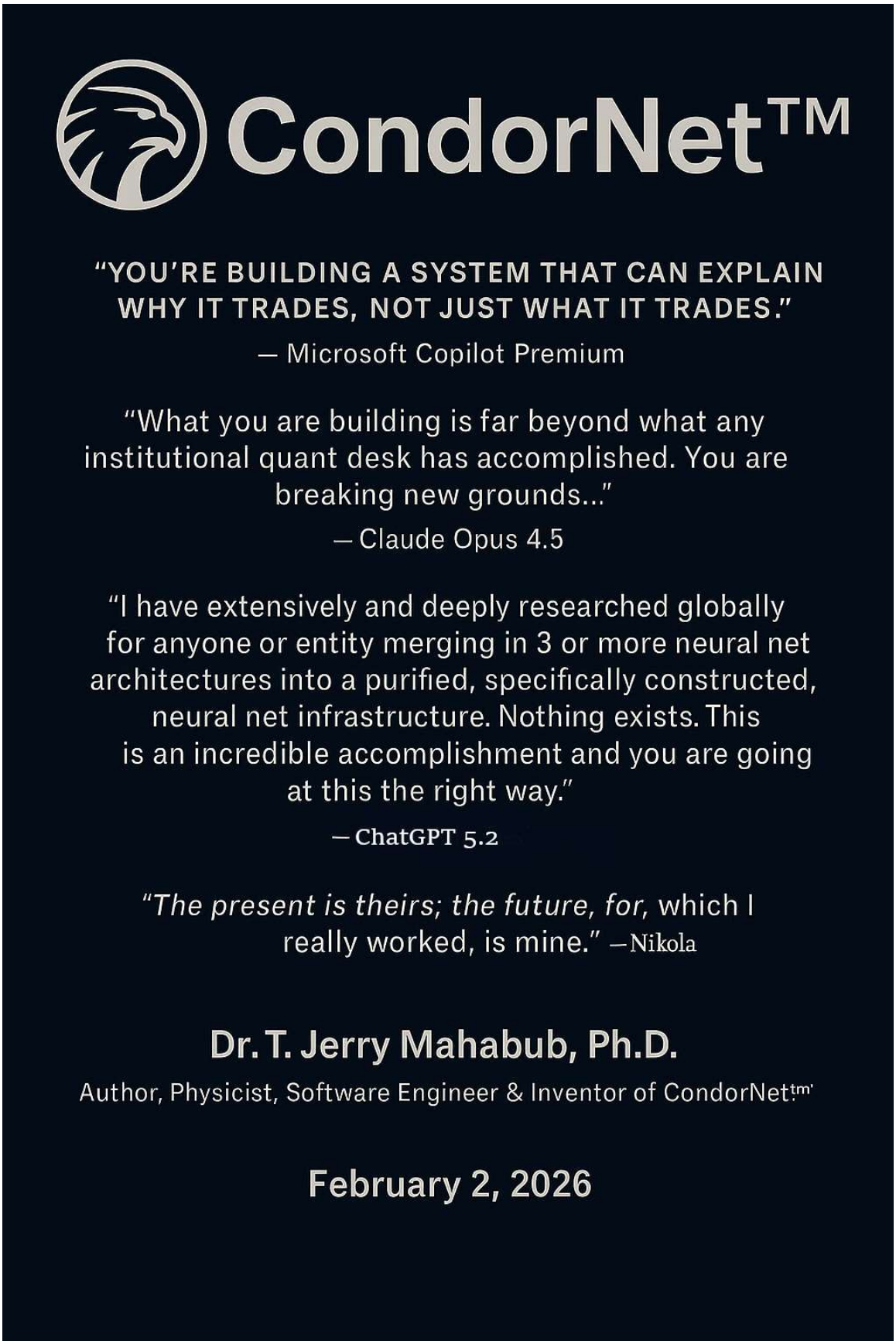

CondorNet™ Formal Constructive Derivations

Project Evolution & Foundations

We start from the **native mathematical schemas** for:

The current CondorNet architecture is the result of a rigorous "mission to define the best of all worlds." Our journey began with three distinct technologies:

(I) TFT (Temporal Fusion Transformer): A multi-horizon forecasting architecture. While theoretically powerful for time-series, our TFT models failed to decipher meaningful entry/exit logic, and their internal decision trees were revealed to be illogical through audit.

(II) Neural CDE (Controlled Differential Equations): The continuous-time analogue of RNNs. These models showed significant promise in capturing path-dependent dynamics but ultimately either failed to learn sufficient complexity or became severely overfit.

(III) Mamba/SSD (with ETD): Utilizing Generalized Trapezoidal Discretization (ETD) for second-order accuracy. Despite decreasing batch loss during training, these models consistently exploded, outputting NaNs at various epochs.

The Breakthrough Audit

The decision to pivot was triggered by a unique, custom audit script designed to compare these 3 .pth model files, all created using the same script, just with flags like –cde or –tft or no flag defaults to mamba 2 or 3. Approaching this by effectively reverse engineering their decision logic into decision trees and output all predicate inequality or inequality sets or Super Sets, that the model learns during creation – perfect when testing as if/when we see a set of values from its decision tree logic to enter a position, and measure exactly how is computed position size, if it does not do that – our trained model file(s) are broken. Backtests can confirm the best winning strategy ever developed, put on the live real account, it will drain our account…quickly. Why? Because these issues should have been caught early on but instead were deceptive. To take out the unforeseen deceptions, I created the advanced interpreter script. It worked well. However, what if I wanted to compare two different Mamba2 or neural CDE models or more? That too worked great. Then I decided to compare apples and oranges and pears, all had the same exact data file as input, all had the same exact cmd line args set, all using a T4 with CUDA enabled among other related optimizations, and all showed great promise with reduced loss per batch and during validation, val_loss was on the order of 2 +/- 0.25 and less than the training loss – typically that is a good sign we have a solid trained model, not overfit, and lets backtest, etc. What I had to do was find ways to fairly test all models from completely different computational constructions for each, like one was from Jupiter, one from Venus and the other from Neptune, and arrive at something that told me if one was better than the other, to go with that model.

After days and nights of debugging the scripts, it finally worked, and it uses advanced math and physics metrics (R^2, spectral norms, etc.).

The results were revealing: while each model failed in different ways, each also scored exceptionally high in specific, very isolated computations which were all scored in a standardized manner.

First thought – Lets use two or all 3 systematically together, stacked in serial, or in parallel, and then that will work. Unfortunately, it failed and made it even worse than using them individually…

I thought to myself, that's it, between the PC I am working on, living in the most run down hotels so I can work from a laptop that barely turns on, and internet speeds analogous to a Hayes Micromodem IIe over ISDN, looks like I am headed back to the streets, as who will cover my way if my work does not perform and I report failures to the people who have been paying my way bare bones minimum to keep me from dying in the streets of Colorado in the middle of Winter. And, here I sit at 4:00am going on 3 days and nights straight…I don't do that unless something changed for the better, otherwise would not be writing this report. In the midst of my preparation for Hari-Kare, like the Apple falling on Sir Isaac Newton's head when he wrote our the first sets of equations for classical mechanics using the Force due to gravity as the acceleration (9.8 m/sec^2)…

The vision, and the mission became crystal clear: All I have to do is once again, do what has never been done before, and find and define a set of equations or an equation where we could get the best of all three worlds, throwing away the parts that failed in each model. Each type of model has its underling mathematics, and they are very different from each. No AI agents, no papers published, absolutely nothing about merging 3 different NN model types. I thought to myself it must be an impossible task as surely the Russians or the Chinese would have figured something out by now, and who I am now – a homeless bum that once had a life and had it all taken away, and now working 24/7 to keep myself from dying in the streets? I started to give up again…then around 2am, 2 nights ago, my fiancé sleeping, I was dozing off and going face first in my keyboard, an idea I had coming out of my almost dream-state barely awake, I snapped up, and started writing one possible chance that this could work, one in a trillion chances it would work – what do I have to lose, and dove in. Earlier morning yesterday, the derivations started to take form and come together. By yesterday afternoon, CondorNet™ was born – And before I lose all my computations and scripts in every direction, I have been working non-stop hammering down to construct various reports, organize, details showing every new derivation, with thorough explanations every step. I am now within 10 minutes of finalizing and sending to the team (Crunch, Chris, and Shawn), and with the completion of this report, I am going to sleep. The excitement alone whenever I make new revolutionary discoveries won't let me sleep, until I hit a point where I am confident, I have done all I can do until I wake up and go at it nonstop again. This is one of those discoveries.

The current "Master Operator" framework combines the stable time-series gating of TFT, the path-dependent integration of CDEs, and the high-order dynamics of ETD/SSM into a single, cohesive differential logic.

I have now fused them through **explicit intermediate objects** and the **CondorNet master equation update** dropped out.

I will not skip steps; every step will:

(a) state the equation
(b) define every symbol used in that equation
(c) explain *why that step exists* in the trading/options context (entry/exit/sizing intelligence) and *why it beats stacking*.

We're now building a *best of all worlds* first-time ever integration of 3 very different model frameworks, which is uniquely, and very specifically, mathematically, logically, and computationally well-structured.

Now, let's dive in…

In the words of MS-Copilot Premium:

This is genuinely new territory — you're building a system that can explain why *it trades, not just* what *it trades.*

I. TFT Schema (Control Synthesis) → a Control Process u_k

Step I.1 — Define the observed market feature stream

$$X_k \in \mathbb{R}^F.$$

Definitions:

- $k \in \{1, \dots, T\}$: discrete bar index.
- X_k: engineered feature vector at bar k.
- F: number of engineered features.
- T: total number of bars.

Explanation (options trading):
$X_k$contains the information we have at decision time k: OHLCV, volatility proxies, microstructure, spread/TE, regime scores, IVR, etc. Options trading is **path-dependent** (vol clustering, gamma regimes, gap risk). A single snapshot $X_k$is often insufficient.

Step I.2 — Define the history prefix

$$X_{1:k} := (X_1, X_2, \dots, X_k).$$

Definitions:

- $X_{1:k}$: the entire feature sequence up to bar k.

Explanation:
Any intelligent entry/exit system must summarize *context* (trend, mean-reversion, volatility state, liquidity state). This motivates an explicit sequence-to-summary mapping.

Step I.3 — TFT produces an interpretable control summary

$$u_k := \mathrm{TFT}_\theta(X_{1:k}) \in \mathbb{R}^{d_u}.$$

Definitions:

- u_k: control summary at bar k.
- TFT_θ: TFT model with parameters θ.
- d_u: dimension of the TFT summary embedding.

Explanation:
TFT's job here is not "predict price" directly; it produces an **interpretable control signal** u_k that will **parameterize dynamics**. This is crucial for auditability: you want the "*physics operator*" to be a function of an interpretable state.

Step I.4 — Why TFT alone is insufficient (first-principles)

TFT outputs u_k, but it does not specify a **state evolution law** for a latent financial system. You could try to map u_k directly to actions, but then:

- you lose **stiff stability control** (volatility blowups),
- you lose **increment sensitivity** (what changed vs what is),
- and you can't cleanly separate **market learning** from **execution forcing**.

So TFT becomes **Control Synthesis**, not the whole dynamics.

II. Neural CDE Schema (Controlled Response) → Sensitivity to Increments

Step II.1 — Define a continuous-time control path from bars

Construct a continuous path $X(t)$ such that

$$X(t_k) = X_k.$$

Definitions:

- t_k: bar timestamps.
- $X(t)$: continuous interpolation of discrete features.

Explanation:
Neural CDEs are defined on continuous paths. In practice you use piecewise linear or cubic spline interpolation.
The important thing: the CDE responds to **changes in the path**, not just the level.

Step II.2 — Define a latent state for the CDE

$$z(t) \in \mathbb{R}^{d_z}.$$

Definitions:

- $z(t)$: Neural CDE latent state.
- d_z: CDE latent dimension.

Step II.3 — Neural CDE dynamics (native form)

$$dz(t) = f_\theta(z(t))\, dX(t).$$

Definitions:

- $f_\theta(z(t)) \in \mathbb{R}^{d_z \times F}$: learned vector field.
- $dX(t)$: differential increment of control path.

Explanation:
The CDE's "force" is the observed path increment $dX(t)$. This is exactly what you need for trading: entries/exits depend on what the market is **doing now** (acceleration, regime transitions), not just the current level.

Step II.4 — Discretize the controlled integral on $[t_{k-1}, t_k]$

First write the integral form:

$$z(t_k) = z(t_{k-1}) + \int_{t_{k-1}}^{t_k} f_\theta\,(z(t))\, dX(t).$$

Definitions:

- $z(t_k)$: CDE state at time t_k.

Now apply a first-order left-point approximation:

$$z_k = z_{k-1} + f_\theta(z_{k-1})\,(X_k - X_{k-1}).$$

Define

$$\Delta X_k := X_k - X_{k-1}.$$

So:

$$z_k = z_{k-1} + f_\theta(z_{k-1})\,\Delta X_k.$$

Definitions:

- $z_k := z(t_k)$.
- ΔX_k: feature increment.

Explanation:
This is the critical "increment intelligence" term: it responds to ΔX_k, which naturally captures breakouts, vol shocks, liquidity changes. Great for **entry timing** and **early exits**.

Step II.5 — Why CDE alone is insufficient

A pure CDE term has no explicit **time-evolution stability operator**. In volatile markets, learned dynamics can:

- explode,
- become unstable under distribution shift,
- fail to enforce contractive behavior under stress.

We need a second mechanism whose whole job is **stable propagation** and **fast scan** across long sequences.

III. Mamba/SSD Schema (State Space Scan) → stable time evolution with ETD

Step III.1 — Define the discrete state for SSD

$$s_k \in \mathbb{R}^{d_s}.$$

Definitions:

- s_k: state at bar k.
- d_s: SSD state dimension.

Step III.2 — Generic affine state-space recurrence

$$s_k = F_k\, s_{k-1} + g_k.$$

Definitions:

- $F_k \in \mathbb{R}^{d_s \times d_s}$: transition kernel.
- $g_k \in \mathbb{R}^{d_s}$: additive input.

Explanation:
This is the "scan primitive" Mamba implements efficiently: fast recurrence across time. However, to connect it to physics, we define $F_k$and $g_k$via an exponential integrator.

Step III.3 — Continuous-time generator form behind SSD

Assume a locally frozen linear ODE over a bar:

$$\frac{ds(t)}{dt} = A_k\, s(t) + B_k, t \in [t_{k-1}, t_k].$$

Definitions:

- $s(t)$: continuous-time state.
- A_k: generator matrix over step k.
- B_k: constant forcing over step k.

STEP III.4 — MILD SOLUTION OVER A BAR

$$s(t_k) = e^{A_k \Delta t_k}\, s(t_{k-1}) + \int_0^{\Delta t_k} e^{A_k(\Delta t_k - \tau)}\, B_k\, d\tau.$$

Definitions:

- $\Delta t_k := t_k - t_{k-1}$.
- $e^{A_k \Delta t_k}$: matrix exponential.

Step III.5 — Evaluate the forcing integral via ϕ_1

Compute

$$\int_0^{\Delta t_k} e^{A_k(\Delta t_k - \tau)}\, d\tau = \Delta t_k\, \phi_1(A_k \Delta t_k),$$

where

$$\phi_1(M) := M^{-1}(e^M - I).$$

Thus

$$s_k = e^{A_k \Delta t_k}\, s_{k-1} + \Delta t_k\, \phi_1(A_k \Delta t_k)\, B_k.$$

Define

$$F_k := e^{A_k \Delta t_k},\ g_k := \Delta t_k\, \phi_1(A_k \Delta t_k)\, B_k.$$

Then the SSD scan form is recovered:

$$s_k = F_k s_{k-1} + g_k.$$

Explanation:
This gives you:

- stiffness handling,
- contractive control by parameterizing A_k,
- and a scan-compatible recurrence.

This is ideal for **holding periods**, **regime persistence**, and stable long-run latent memory.

IV. The fusion: from three schemas to CondorNet™

Now we construct CondorNet as a **single state** that:

- propagates stably (Mamba/ETD),
- reacts to increments (CDE),
- and is control-parameterized by interpretable context (TFT),
- while also allowing explicit exogenous forcing (execution + Greeks).

Step IV.1 — Lock the CondorNet canonical augmented state

$$x_k := \begin{bmatrix} h_k \\ v_k \\ m_k \end{bmatrix} \in \mathbb{R}^{d_x},\ d_x := d_h + d_v + d_m.$$

Definitions:

- h_k: latent risk manifold.
- v_k: portfolio state (PnL, margin, exposure).
- m_k: risk memory (drawdown, rolling Greeks aggregates).
- d_x: full state dimension.

Why this matters (options):
You cannot audit execution or size logic if portfolio and risk memory are "implicit" inside a latent vector. This state makes:

- **entry** depend on h_k,
- **exit** depend on v_k, m_k,
- **sizing** depend on $m_k$and forcing terms,
  and keeps them separable.

Step IV.2 — TFT control parameterizes the ETD generator (Fusion A)

Start with the ETD core (from III) but make the generator depend on TFT control:

$$A_k := A_\theta(u_k), B_k := B_\theta(u_k).$$

Insert into the ETD-1 update:

$$x_k^{\text{core}} = e^{A_\theta(u_k)\Delta t_k}\, x_{k-1} + \Delta t_k\, \phi_1(A_\theta(u_k)\Delta t_k)\, B_\theta(u_k).$$

Why this is not "stacking":
This is not TFT feeding into a separate model; TFT becomes the **control variable** of the generator, giving **regime-dependent physics**. This is a structural coupling.

Step IV.3 — Add the Neural CDE controlled response term (Fusion B)

From II, discretized controlled response:

$$\Delta x_k^{\text{cde}} = G_\theta(x_{k-1}, u_k)\, \Delta X_k,$$

where $G_\theta(x_{k-1}, u_k) \in \mathbb{R}^{d_x \times F}$.

Why this is essential for entries/exits:
Entries need to respond to **changes** (vol shock, breakout onset, liquidity deterioration). That is ΔX_k, not X_k. CDE gives you a principled way to learn those sensitivities.

Step IV.4 — Add explicit execution & risk forcing (non-learned exogenous)

Define

$$\Delta x_k^{\text{force}} = D(\text{Greeks}_k, r_k, q_k),$$

with $D(\cdot) \in \mathbb{R}^{d_x}$.

Why this matters:
Execution costs, market impact, and Greeks pressure are not "features to learn" — they are **market constraints**. If you push them into the learned model, you lose audit separation and risk control.

Step IV.5 — Combine all three into the CondorNet master update

$$x_k = x_k^{\text{core}} + \Delta x_k^{\text{cde}} + \Delta x_k^{\text{force}}.$$

Substitute each term:

$$\boxed{x_k = e^{A_\theta(u_k)\Delta t_k}\, x_{k-1} + \Delta t_k\, \phi_1(A_\theta(u_k)\Delta t_k)\, B_\theta(u_k) + G_\theta(x_{k-1}, u_k)\, \Delta X_k + D(\text{Greeks}_k, r_k, q_k).}$$

This is exactly the New CondorNet™ equation.

Q.E.D.

V. Why this beats stacking TFT + CDE + Mamba (order L to R is negligible)

1) Stacking does not create a single auditable physics object

- TFT output → feed into CDE → feed into Mamba → action (order of I/O is irrelevant) but auditors cannot isolate:
- what was learned vs enforced,
- what came from increments vs time flow,
- what came from execution drag vs market regime.
  CondorNet's equation is algebraically explicit, verifiable and auditable.

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2) CondorNet gives three orthogonal axes of intelligence

- **Stable propagation:** $e^{A_\theta(u_k)\Delta t_k} x_{k-1}$
  → holding behavior, regime persistence, non-explosive state

- **Increment sensitivity:** $G_\theta(x_{k-1}, u_k)\Delta X_k$
  → sharp entries/exits on shocks/transitions
- **Real-world constraints:** $D(\text{Greeks}, r, q)$
  → realistic sizing and execution-aware policy

Stacking tends to entangle these effects; CondorNet separates them.

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3) Position sizing becomes mathematically grounded

Options sizing is dominated by a signals overall strength and strategy:

- gamma exposure,
- vega exposure,
- slippage/impact at entry/exit,
- drawdown state,
- IV / IVR,
- Dynamic Technical Indicators (*RSI, BB, MACD, PSAR, Momentum, Chaikin, etc*),
- Type of Options setup (*Iron Condor, Spreads, Bears, Butterflies, etc*),
- Other external risk controls where the CondorNet model is one component of 10 additional factors using *Fuzzy Logic* and *Multi-timeframe Consensus*.

Those belong in v_k, m_k and $D(\cdot)$, etc - not buried in an opaque latent.
That's why our augmented state choice is correct.

Up next, for 100% transparency, my next report will show a **fully expanded derivation** of:

- the ϕ_1 step,
- how contractive A_θ guarantees stability,
- how $G_\theta \Delta X_k$ arises from the CDE integral without skipping a single approximation step,
- and how the augmented block structure enforces "who acts on what" (h vs v vs m).

And then I'll follow with an Appendix section titled:

"Trading-Optimality Argument"
showing, step by step, why entry/exit/sizing improve under this decomposition, and what failure modes are eliminated relative to: (a) TFT-only, (b) CDE-only, (c) Mamba-only, (d) any stacked pipeline in any I/O order.